

Sales Analysis Report

Executive Summary

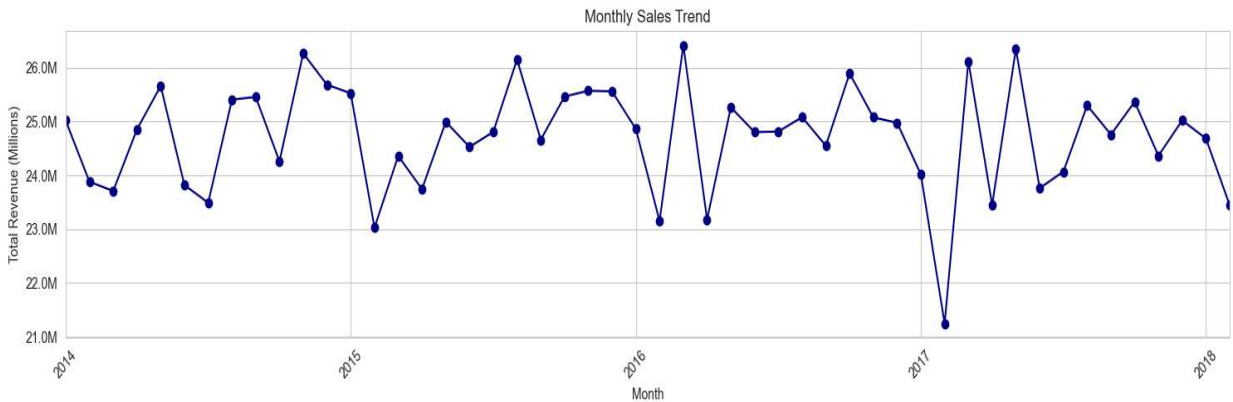
The notebook analyzes 64,104 sales-order records across customers, products, channels, regions, and states. It derives revenue, cost, profit, and profit-margin metrics and explores performance trends across multiple business dimensions.

Data Overview

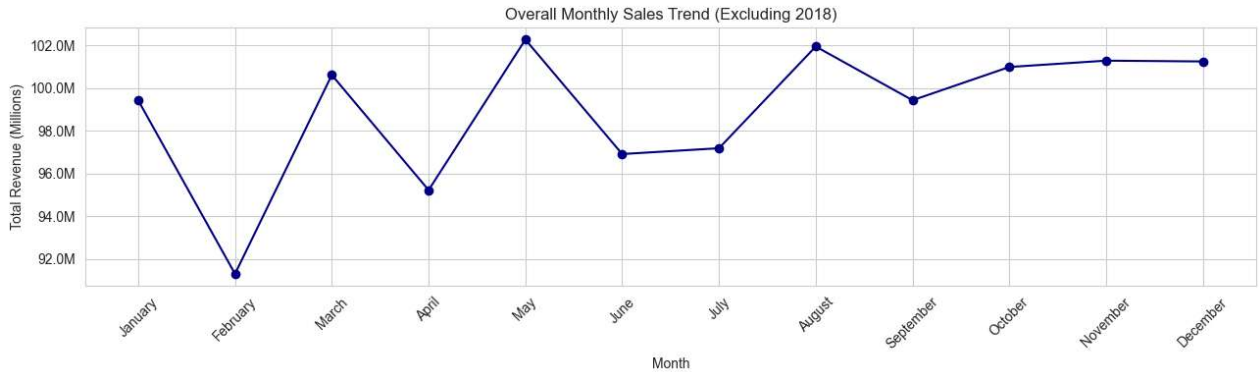
Datasets include Sales Orders, Customers, Products, Regions, State Regions, and 2017 Budgets. The workflow merges these sources, cleans fields, calculates profit metrics, and prepares data for reporting.

Recommended Charts Included in the Analysis

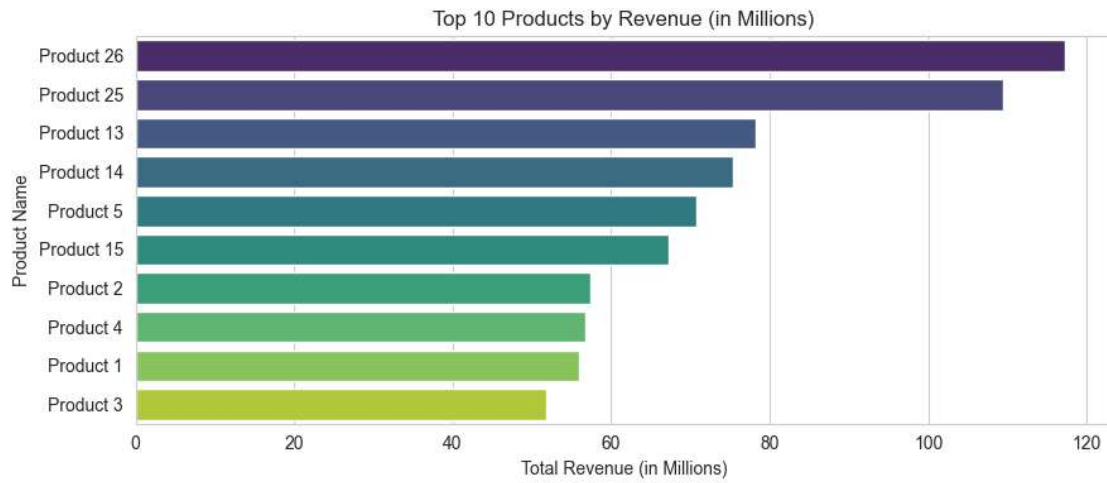
- Monthly Sales Trend



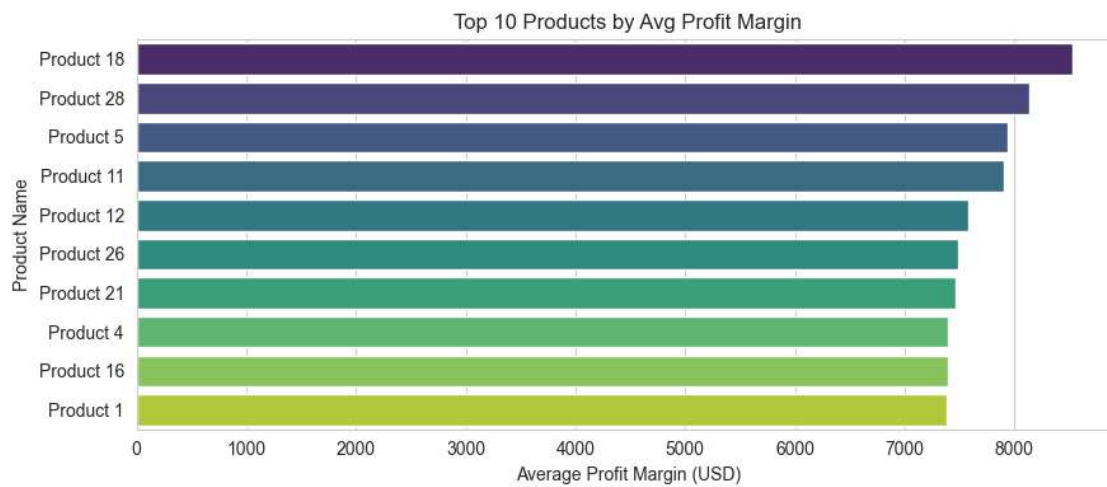
- Seasonality by Month



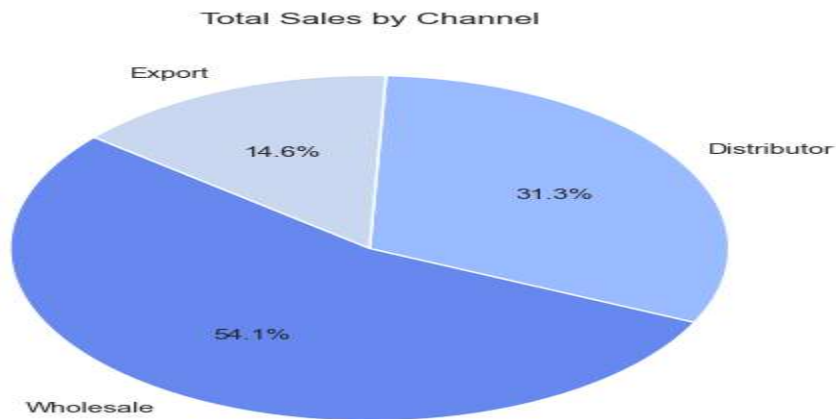
- Top 10 Products by Revenue



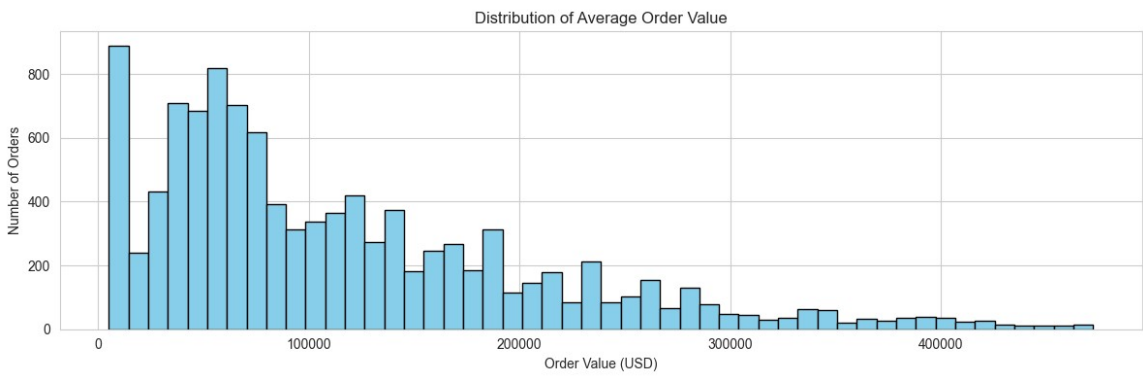
- Top 10 Products by Average Profit



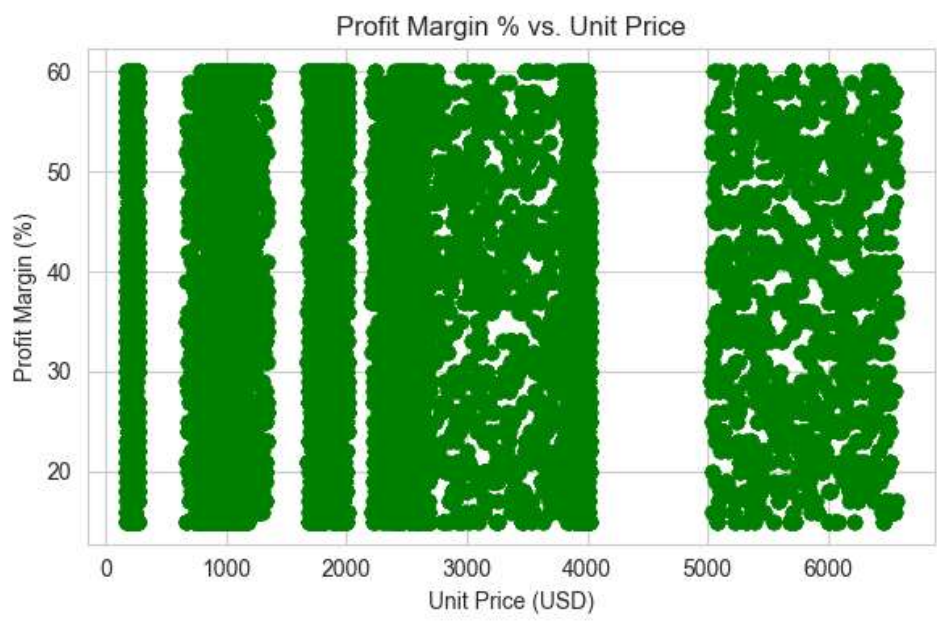
- Sales by Channel (Pie Chart)



• Order Value Distribution



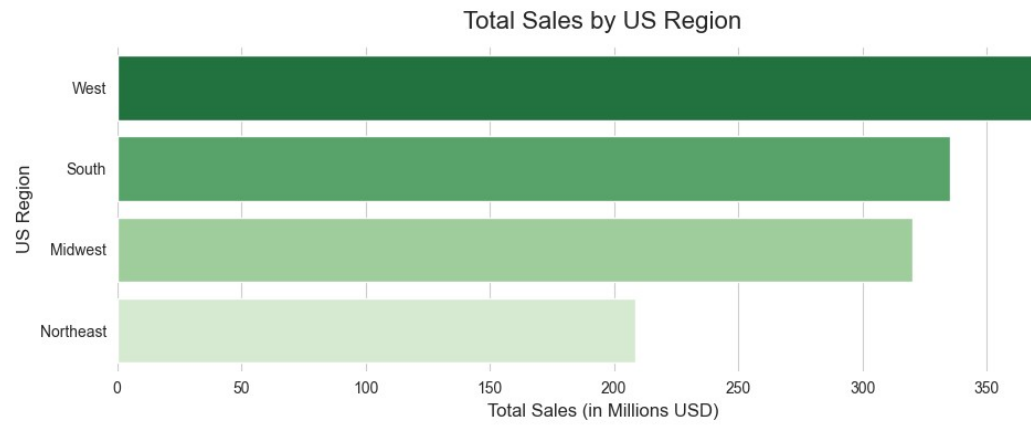
• Profit Margin vs Unit Price (Scatter Plot)



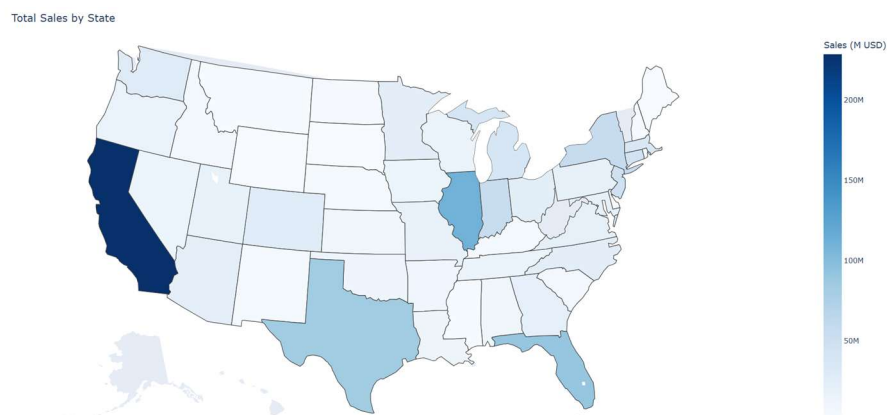
• Product Price Distribution (Box Plot)



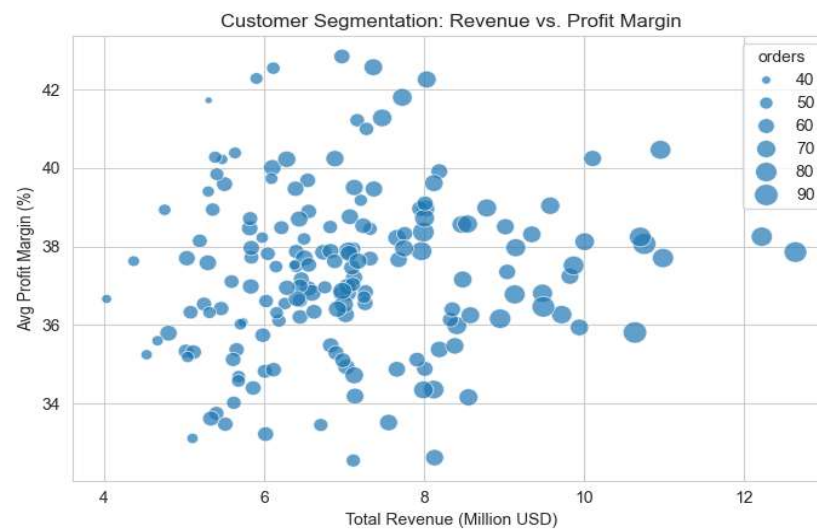
• Sales by US Region



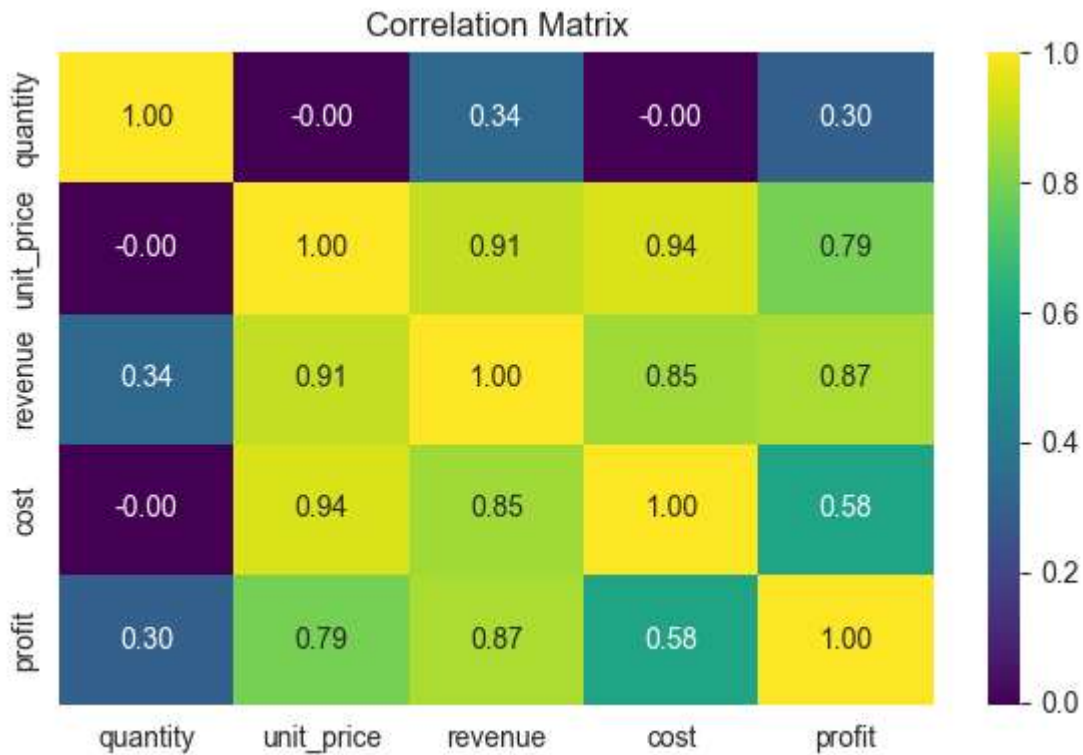
- Top States by Revenue and Orders



- Customer Segmentation Bubble Chart



- Correlation Heatmap



Key Business Insights Template

Populate this section with actual chart findings after rerunning the notebook with the source dataset. Typical insights include top-performing products, highest-margin channels, strongest regions, seasonality trends, customer concentration, and revenue-profit relationships.

Recommendations

Focus investment on high-revenue/high-margin products, improve low-performing channels, monitor regional performance, and use customer segmentation to prioritize strategic accounts.